

Next Word Prediction Using Deep Learning : A Comparative Study

Milind Soam

Department of Computer Science and Engineering
Amity University, Uttar Pradesh, India
milind.soam1@s.amity.edu

Dr. Sanjeev Thakur

Department of Computer Science and Engineering
Amity University, Uttar Pradesh, India
sthakur3@amity.edu

Abstract-- Deep learning is a subclass of machine learning, it mimics the functionality of the human brain the way it processes and creates pattern in the facts for choice making . It is basically an AI function that has networks capable of learning unsupervised data that is shapeless. The succeeding word forecast is performed on dataset consisting of texts. Next Word Prediction is an application of NLP (Natural Language Processing) . It is also known as Language Modelling .Basically it is the process of predicting the next word in a sentence. It has many applications which are used by most of us such as auto-correct which is mostly used in Emails / Messages ; it has also its usage in MS Word or google search where forecasts the next word based on our search history or the search we did for globe. In this work we have studied NLP , different deep learning techniques such as LSTM, BiLSTM and performed a comparative study. We found satisfactory results in BiLSTM and LSTM [1]

The accuracy received using BiLSTM and LSTM are : 66.1 % and 58.27% respectively.

Keywords—Deep Learning , Recurrent Neural Network , LSTM , BiLSTM

I. INTRODUCTION

Artificial Neural Networks are part of the deep learning that are inspired by the human neurons. Because of ANN there were significant milestones in all the fields resulting in bringing evolution to those fields .These milestones were achieved because it is the backbone of deep learning.

Deep learning is a subset of machine learning and it mimics the functionality of the human brain the way it processes the data and creates pattern for the use in decision making . It is basically an AI function that has networks capable of learning unsupervised data that is unstructured or unlabelled. Artificial neural networks are basically the backbone of deep learning which led to the achievement of bigger milestones in this field of machine learning where they brought evolution on how to approach the

problems, therefore the deep learning has an important role to play in machine learning and artificial Intelligence.

The next word prediction on Medium Articles dataset is a good starting project in the field of NLP and deep learning in the area of next word prediction.

Next word prediction is an important part of finding or predicting the reviews or search query or search results from view point of customers or working people or internet users based on their search history , typing history or globally used words with specific sentence. Next word prediction is a technique that is used to determine or predict the next words and the data over here is the medium articles dataset . Next Word Prediction is a natural language processing technique. Next word prediction is often performed on textual data that helps people to autocomplete or explore other search queries and on that basis improve the standard of quality or the standard of services that is given to the customer or people either on the web or on the platforms or apps that support next word prediction therefore next word prediction in today's world has an important role to play and has become an essential tool.

II. LITERATURE REVIEW

Radhika Sharma et al. [1] used deep learning models implemented using BiLstm and lstm technique.. This study focuses on getting a novel method for the word prediction in Hindi Language. Each text data would be cleaned then the dictionary of unique words would be created by splitting the words in the dataset and filters out the unique words. The dataset used for this research was developed by IIT Bombay at the Indian Language Technology center.[2]. The accuracy achieved using both LSTM and BiLSTM is 59.46% and 81.07% respectively.

Zabit Hameed et.al[3] presents an efficient deep learning model in their paper which aims at the classification of binary sentiments on three datasets including IMDB Dataset to classify the polarity of the sentiments that are in the form of peoples opinions attitudes and emotions expressed in written text. this paper uses a word embedding technique known as glove vector of 300 dimensions[4] . The

deep learning technique used here is bidirectional long short term memory(BiLSTM) with RMSProp optimizer .The results was that BiLSTM achieved an accuracy of 90.585% on IMDb dataset.

Saikirin Gogineni et.al[5] came up with automatic text categorization, which is a key task in today's digital environment for analysing or classifying the user's evaluations and feelings without human interaction. It would assist users in making better movie choices. To measure the effectiveness of the models, the study employs different metrics such as accuracy precision recall. The authors employed RNNs such as LSTM, Tf-idf, and GRU, as well as Word2Vec[6] for improved word embedding approaches. The efficiency of traditional machine learning algorithms and deep learning models has been compared in this article, and improved models or efficient architecture has been offered.

Omor Faruk Rakib et.al[7] performed Next word Prediction and sentence prediction in Bangla Language using GRU on n-gram dataset to create such models. The work has been compared with LSTM model on n-gram dataset and Naïve Bayes with Latent Semantic analysis. The dataset is in Bangla Language gathered from different sources and is gathered around 170 thousand .The proposed model achieved an accuracy of 99.70% for 5-gram model , 99.24% for 4-gram , 95.84% for tri gram , 78.15% and 32.17% for bigram and unigram models respectively.

Younggi Kim et.al[21] paper on an activity Embedding approach for next activity prediction in a multi-user smart space used word embedding algorithm that maps words to vectors. After the vectorization is performed the same is trained using LSTM model to predict a single vector coordinate for next activity from a group of previously occurred activities. The accuracy achieved by this model is 82%.

Jingyun Yang et. Al[22] worked on natural language Word Prediction Model based on Multi Window convolution and residual network where the authors proposed MCNN-ReMGU model for natural word predictions The results were output by using the activation function Softmax .The experiment was performed on Penn Treebank and WikiText-2 datasets .The best results were obtained using MCNN-ReMGU using 150 hidden units and achieving an accuracy of 95.2%.

YongfengDong et.al[8] paper introduced a capsule network model named caps by LSTM for sentiment analysis on IMDb data set and other different data sets. A convolutional layer is used in the beginning

of the caps BiLSTM to transform the instance to hide vector. The module then constructs an n-gram model. The capsule model is used to calculate the state probability of a certain capsule. A higher coupling coefficient is assigned if any given instance has a large state probability among other capsules. The output of capsule enters into a BiLSTM structure to fusion the data features. This model achieved better performance than the CNN , LSTM and BiLSTM model.The author used Glove Embedding of dimension 300[4].

Md. Rakibul Haque et.al[9] performed performance analysis of different neural network on the IMDb Dataset. The performance was based on deep learning techniques as LSTM, CNN ,and a hybrid model of LSTM-CNN. The word embeddings that was used was provided by Keras which contains 8000unique words. The results were LSTM and CNN achieved an accuracy of 88% and 90% respectively. The hybrid model achieved an accuracy of 89%. The CNN outperformed them both.

Saeed Mian Qaisar[10] did the same work of performing sentiment analysis on IMDb Dataset by using RNN Deep Learning . The analysis was performed by using LSTM with Doc2Vec and achieved an accuracy of 89.9%.

III. METHODOLOGY

A. Dataset

The Dataset used is the Medium articles dataset[11] which is the largest dataset available on Medium articles. The dataset contains 6508 articles. The dataset can be available from Medium website or it can be downloaded from Kaggle[2]. The dataset has been trained on categorical crossentropy loss and adam optimizer with an activation function set on softmax.

B. Long Short Term Memory

LSTM is a version of RNN which is a sequential network that allows information to be persistent. It solves the problem of vanishing gradient in RNN . Basically RNN is used for persistent memory. As humans remember what is going to happen further when watching a video or while reading a book they know what would be happening further ; similarly RNNs also work in the same way and remember the previous information to use it for processing the current input. the major drawback of RNN is that because of the vanishing gradient they

cannot remember long term dependencies. and so does lstm is designed to avoid this problem.

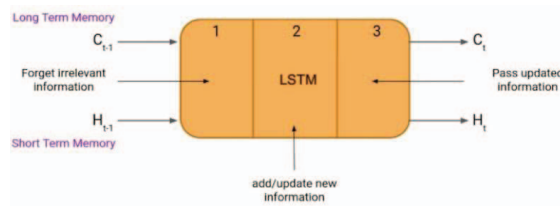


Figure 1 LSTM Architecture[14]

From the figure1 we can see that there are 3 parts of LSTM.[14] The first part chooses whether to remember the information from previous timestamp or not . If in case it finds the information irrelevant it can forget that information. The second block learns from the current input or basically it tries to learn new things from the current input. The third part chooses to update the information from current timestamp and passes it to the next timestamp. These three parts are known as gates of LSTM. The first part of this gate is known As forget gate , second part as input gate and the third part as output gate

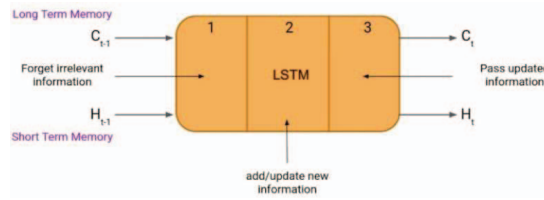


Figure 2 LSTM Layers[15]

Just like RNN LSTM also has hidden states as depicted in Figure2[15]. $H(T-1)$ is known as the hidden state of previous timestamp and H_T is known as the hidden state of current timestamp. LSTM also has a cell state known as $C(T-1)$ for previous timestamp and C_T for current timestamp. Short term memory is known as hidden state, whereas long term memory is known as cell state.

One LSTM layers are used with each of them having 150 units required in output and a dense layer of output unit 1 which uses softmax activation function[16][17].

Adam optimizer is used.[19]

The LSTM layer is used at a learning rate or gradient descent[18] of 0.01.

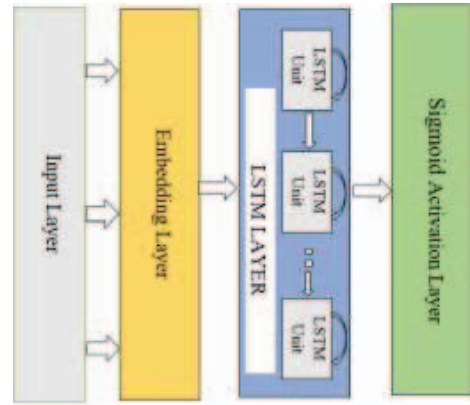


Figure 3 The working of LSTM Network[14]

In Figure 3 we can observe the working of the LSTM Layer[14].The Input Layer is present followed by the embedding layer . After the embedding is done the output passes through LSTM layer where the deep neural network that is LSTM performs it's task of training and finally the output is passed through the Dense Layer having activation function as softmax which is responsible for generating the output that is a vector of all probabilities.

C. Bi-LSTM

BiLSTM or bidirectional LSTM is version of RNN and are improved version of LSTM. It is designed to improve the model's performance in sequence classification problems On the input sequence, it trains two LSTMs rather than one LSTM. The input sequence is used for the first training, and a reversed duplicate of the input sequence is used for the second. It leads to quicker and more thorough learning of the problem, as well as providing new context to the network.. The primary difference between BiLSTM and LSTM is that LSTM will maintain information from input that has previously gone through its hidden state, whereas bidirectional will run the inputs in two directions, one from the past to the future and the other from the future to the past. In a backwards-running LSTM, information from the future is only preserved, but by combining the two hidden states, information from the past and future may be preserved at any point in time.

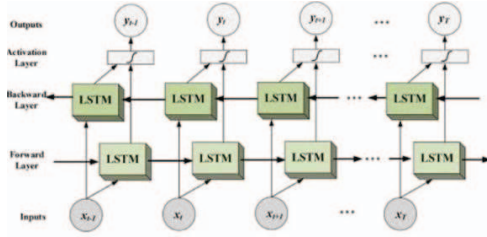


Figure 4 BiLSTM Network[20]

In Figure 4 we can see the Forward and Backward Layer in BiLSTM that make the BiLSTM Network.[20].

The same methodology is applied in BiLSTM as is applied in LSTM with the exception of using one Bidirectional layers of same unit size as LSTM . The Dense Layer is the same as in LSTM. The number of epochs are 150 for both the models during training.

IV. EVALUATION

The evaluation of the model relies on the confusion matrix , the graph it plots and the evaluation metrics that are precision , recall , accuracy and F1 Score.

The most general evaluation metric is the accuracy and loss graph .

The experiment was performed on Google Colab where the Runtime Environment was set to GPU. The machine that is used for the performing of this experiment has specifications:

RAM : 8GB

Processor : Processor Intel(R) Core(TM) i5- 7200U
CPU @ 2.50GHz, 2701 Mhz, 2 Core(s), 4

Logical Processor(s).

GPU : AMD Radeon™ R5 M330

VRAM : 2048 MB

The various evaluation metrics have been defined below.

$$\text{Accuracy} = \frac{TP+TN}{TP+FP+TN+FN} * 100$$

$$\text{Precision} = \frac{TP}{TP+FP} * 100$$

$$\text{Recall} = \frac{TP}{TP+FN} * 100$$

$$\text{F1 score} = \frac{2}{\frac{1}{\text{Recall}} + \frac{1}{\text{Precision}}} * 100$$

Recall Precision

The ratio of the total number of correctly categorised cases to the total number of occurrences is known as accuracy. Precision is defined as the model's ability to effectively determine positive cases. The model's recall shows us how often it correctly predicts favourable events. When both Precision and Recall are important, the F1 score comes in handy. It's the sum of both entities' harmonics.

Figure 5 shows the Graph that is plotted on the basis of 'Model Accuracy' and 'Model Loss'.

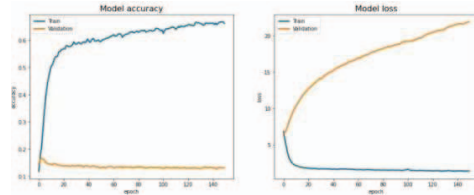


Figure 5 Model Accuracy and Model Loss for LSTM

The Model Accuracy graph is created for accuracy vs epochs whereas Model Loss is created for loss vs epoch.

For the BiLSTM model we can see from Figure 6 it's accuracy and loss graph that is generated when the model was trained.

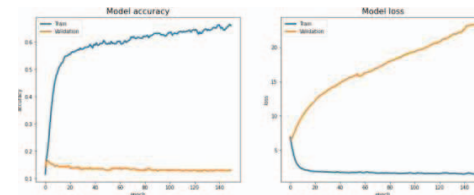


Figure 6 Model Accuracy and Model Loss for BiLSTM

From Figure 6 we can observe that the graph is plotted for model accuracy and model loss.

The Model Accuracy graph is created for accuracy vs epochs whereas Model Loss is created for loss vs epoch.

A. Result analysis

After performing the experimentation we can see from table 1 the analysis of the results.

Model	Accuracy	Loss	Validation_accuracy	Validation_loss
LSTM	58.27	1.74	14.1	13.81
BiLSTM	66.1	1.521	12.83	23.36

Table 1 Value of Accuracy, Loss, Validation accuracy and loss

for each model

The results show that BiLSTM performed the best followed by LSTM giving accuracy of 66.1% and

58.27% respectively . In terms of Loss BiLSTM

has lowest loss .

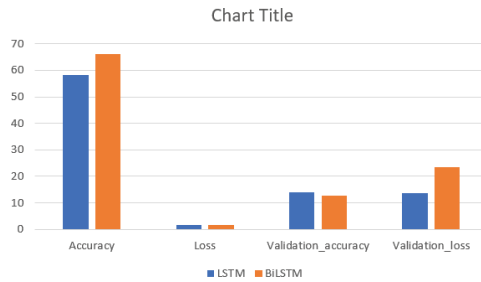


Figure 7 Analysis of the Models

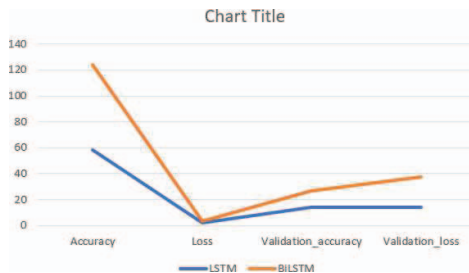


Figure 8 Analysis of the Models

From the figure 7 and 8 it is clearly evident how each model performed and the graph is plotted by taking into account 'Accuracy', 'Loss', 'Validation Accuracy' and 'Validation Loss' for each model.

B. Comparison with related work

The models on which this experimentation is performed achieved higher accuracies for both LSTM and BiLSTM models than the same models used for English Language. Changes in Hyper Parameters were done to obtain better results. These were the best tunings in hyper parameters that resulted such impressive results.

V. CONCLUSION

In this work, we have performed a comparative study and obtained the results as shown in Table 1 and also the same is mirrored in the Bar Chart in figure 7. The conclusion is that BiLSTM performed the best. The result that BiLSTM model performed better was expected as BiLSTM was introduced to solve the drawbacks that occurred in LSTM model. At the end we can see that with BiLSTM we achieved the highest accuracy as well as had the least loss.

We can see that there is high overfitting as is evident from the graphs in figures 5 and 6 for LSTM and BiLSTM respectively. Hence it also

tells that taking more datasets might solve this problem so that training is done on a large number of datasets.

VI. FUTURE WORK

The work would be carried out in the future with now the focus on capsule BiLSTM and hybrid version of CNN and LSTM. Other area of study would be to apply BERT technique in NLP and how these models would be behaving on not just on Medium articles but other datasets of different areas. Also a large dataset would be taken in the future not to just only predict the next word but also to predict what words would be coming for completing the sentences and automatically completing the search queries or sentences.

REFERENCES

- [1] R. Sharma, N. Goel, N. Aggarwal, P. Kaur, and C. Prakash, "Next Word Prediction in Hindi Using Deep Learning Techniques," in 2019 International Conference on Data Science and Engineering (ICDSE), Patna, India, Sep. 2019, pp. 55–60. doi: 10.1109/ICDSE47409.2019.8971796
- [2] A. Kunchukuttan, P. Mehta, and P. Bhattacharyya, "The IIT Bombay English-Hindi Parallel Corpus," arXiv:1710.02855 [cs], May 2018, Accessed: Jan. 22, 2022. [Online]. Available: <http://arxiv.org/abs/1710.02855>
- [3] Z. Hameed and B. Garcia-Zapirain, "Sentiment Classification Using a Single-Layered BiLSTM Model," in IEEE Access, vol. 8, pp. 73992–74001, 2020, doi: 10.1109/ACCESS.2020.2988550.
- [4] Thanakom Sangnetra, "glove.6B.300d.txt", <https://www.kaggle.com/thanakomsn/glove6b300dtxt>, accessed : 2021-05-08.
- [5] S. Gogineni and A. Pimpalshende, "Predicting IMDB Movie Rating Using Deep Learning," in 2020 5th International Conference on Communication and Electronics Systems (ICCES), Coimbatore, India, Jun. 2020, pp. 1139–1144. doi: 10.1109/ICCES48766.2020.9137994.
- [6] M. Tan and Q. V. Le, "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks," arXiv:1905.11946 [cs, stat], Sep. 2020, Accessed: Jan. 22, 2022. [Online]. Available: <http://arxiv.org/abs/1905.11946>
- [7] O. F. Rakib, S. Akter, M. A. Khan, A. K. Das, and K. M. Habibullah, "Bangla Word Prediction and Sentence Completion Using GRU: An Extended Version of RNN on N-gram Language Model," in 2019 International Conference on Sustainable Technologies for Industry 4.0 (STI), Dhaka, Bangladesh, Dec. 2019, pp. 1–6. doi: 10.1109/STI47673.2019.9068063.

- [8] Y. Dong, Y. Fu, L. Wang, Y. Chen, Y. Dong, and J. Li, "A Sentiment Analysis Method of Capsule Network Based on BiLSTM," *IEEE Access*, vol. 8, pp. 37014–37020, 2020, doi: 10.1109/ACCESS.2020.2973711.
- [9] Md. R. Haque, S. Akter Lima, and S. Z. Mishu, "Performance Analysis of Different Neural Networks for Sentiment Analysis on IMDb Movie Reviews," in 2019 3rd International Conference on Electrical, Computer & Telecommunication Engineering (ICECTE), Rajshahi, Bangladesh, Dec. 2019, pp. 161–164. doi: 10.1109/ICECTE48615.2019.9303573.
- [10] S. M. Qaisar, "Sentiment Analysis of IMDb Movie Reviews Using Long Short-Term Memory," in 2020 2nd International Conference on Computer and Information Sciences (ICCIS), Sakaka, Saudi Arabia, Oct. 2020, pp. 1–4. doi: 10.1109/ICCIS49240.2020.9257657.
- [11] Medium Article Dataset
https://www.kaggle.com/dorianlazar/medium-articles-dataset?select=medium_data.csv.
- [12] adityajn105. "GLOVE 6B 50D Word Embeddings",
https://www.kaggle.com/adityajn105/glove6b50_d,
accessed : 2021-05-08.
- [13] J. Pennington, R. Socher, and C. Manning, "Glove: Global Vectors for Word Representation," in *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Doha, Qatar, 2014, pp. 1532–1543. doi: 10.3115/v1/D14-1162.
- [14] SHIPRA SAXENA, "Introduction to Long Short Term Memory (LSTM)",
<https://www.analyticsvidhya.com/blog/2021/03/introduction-to-long-short-term-memory-lstm/>,
accessed : 2021-05-08.
- [15] SHIPRA SAXENA, "Introduction to Long Short Term Memory (LSTM)",
<https://www.analyticsvidhya.com/blog/2021/03/introduction-to-long-short-term-memory-lstm/>,
accessed : 2021-05-08.
- [16] HuiLiu, "Rail transit channel robot systems", <https://doi.org/10.1016/B978-0-12-822968-2.00007-3>.
- [17] S. Sharma, S. Sharma, and A. Athaiya, "ACTIVATION FUNCTIONS IN NEURAL NETWORKS," *IJEAST*, vol. 04, no. 12, pp. 310–316, May 2020, doi: 10.33564/IJEAST.2020.v04i12.054.
- [18] A. Senior, G. Heigold, M. Ranzato, and K. Yang, "An empirical study of learning rates in deep neural networks for speech recognition," in 2013 IEEE International Conference on Acoustics, Speech and Signal Processing, Vancouver, BC, Canada, May 2013, pp. 6724–6728. doi: 10.1109/ICASSP.2013.6638963.
- [19] D. P. Kingma and J. Ba, "Adam: A Method for Stochastic Optimization," *arXiv:1412.6980 [cs]*, Jan. 2017, Accessed: Jan. 22, 2022. [Online]. Available: <http://arxiv.org/abs/1412.6980>
- [20] Matt Cruickshank, "What does it mean by Bidirectional LSTM?".
<https://morioh.com/p/df9310bb0277>, accessed : 2021-05-08.
- [21] Y. Kim, J. An, M. Lee, and Y. Lee, "An Activity-Embedding Approach for Next-Activity Prediction in a Multi-User Smart Space," in 2017 IEEE International Conference on Smart Computing (SMARTCOMP), Hong Kong, China, May 2017, pp. 1–6. doi: 10.1109/SMARTCOMP.2017.7946985.
- [22] J. Yang, H. Wang, and K. Guo, "Natural Language Word Prediction Model Based on Multi-Window Convolution and Residual Network," *IEEE Access*, vol. 8, pp. 188036–188043, 2020, doi: 10.1109/ACCESS.2020.3031200.